ZJU-UIUC INSTITUTE
Zhejiang University-University of Illinois Urbana-Champaign Institute
浙江大学伊利诺伊大学
厄巴纳香槟校区联合学院

Curriculum of Civil Engineering for Class of 2027

Zhejiang University/University of Illinois Urbana-Champaign Institute

(ZJU-UIUC INSTITUTE)

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1. Overview

The Zhejiang University/University of Illinois Urbana-Champaign Institute (ZJU-UIUC Institute) offers a joint dual-degree engineering program. Upon successful completion of the program, and after meeting the graduation requirements of both universities, students will obtain bachelor's degrees separately from Zhejiang University (ZJU) and the University of Illinois Urbana-Champaign (UIUC).

Civil engineering is a profession that applies the basic principles of science in conjunction with mathematical and computational tools to solve problems associated with developing and sustaining civilized life on our planet. Civil engineering works are generally one-of-a-kind projects; they are often grand in scale; and they usually require cooperation among professionals of many different disciplines. The completion of a civil engineering project involves the solution of technical problems in which uncertainty of information and myriad non-technical factors often play a significant role. Some of the most common examples of civil engineering works include bridges, buildings, dams, airports, highways, tunnels, and water distribution systems. Civil engineers are concerned with flood control, landslides, air and water pollution, and the design of facilities to withstand earthquakes and other natural hazards, in addition to protecting our environment for a sustainable future.

The Civil Engineering program comprises seven main areas (construction engineering and management, construction materials engineering, environmental engineering, geotechnical engineering, environmental hydrology and hydraulics, structural engineering, and transportation engineering) and three cross-cutting programs (sustainable and resilient infrastructure systems; energy, water, and environmental sustainability; and societal risk management). Although each area has its own special body of knowledge and engineering tools, they all rely on the same fundamental core principles. Civil engineering projects often draw on expertise from many of these areas and programs.

Our Education Objectives are to educate Civil Engineering students to:

1) Keep exploring new techniques and new knowledge for the sustainable development of human society.

2) Successfully enter the civil and environmental engineering profession as practicing engineers and consultants with prominent companies and organizations in diverse areas that include structural, transportation, geotechnical, materials, environmental, and hydrologic engineering; construction management; or other related or emerging fields.

3) Pursue graduate education and research at major research universities in civil and environmental engineering, and related fields.

4) Pursue professional licensure and advance to leadership positions in the profession.

5) Engage in continued learning through professional development.

6) Participate in and contribute to professional societies and community services.

During their junior and senior years, students are encouraged to take courses and involve themselves in research projects in cross-disciplinary areas with Electrical Engineering, Mechanical Engineering, Computer Engineering, and others.

2. Graduation Requirement

2.1 Graduation Requirements for UIUC degree

2.1.1 Grade Point Average Requirement

A student must maintain a minimum GPA of 2.0 (A=4.0) to remain in good standing and graduate. UIUC degree is subjected to the final review by UIUC.

2.1.2 Junior Eligibility Requirement

To be eligible to enroll in the CEE courses listed in the third year of the curriculum, a student must have completed, with a combined 2.25 grade point average, the mathematics, physics, computer science, and electrical and computer engineering courses listed in the first two years.

2.1.3 Curriculum Requirement

The curriculum leading to the degree of Bachelor of Science in Civil Engineering at UIUC requires **128** academic credits (or academic units, AUs) and is organized into required courses, technical electives, liberal education, and other electives.

i). Required courses, see section 3 for details.	Academic Credit
a) Orientation and Professional Development	5
b) Foundational Mathematics and Science	35
c) Technical Core	25
d) Composition	8
e) Advanced Composition*	4
Total Required AUs:	**77**

ii). Elective courses, see section 4 for details.	
a) Science Electives	3
b) Technical Electives	34
c) Liberal education	12
d) Free Electives	6
Total Required AUs:	**55**

**Students take CEE 300 to satisfy advance composition and technical elective requirement, and credits can be given for both.*

2.1.4 Additional requirements

For UIUC degree, in addition to the specific course and grade point average requirements listed above, each candidate for a bachelor's degree from UIUC must meet the following requirements:

i). Residency Requirement

Earn a minimum sixty (**60**) academic credits of UIUC credit, of which at least twenty-one (**21**) AUs must be 300 or 400 level UIUC credit courses.

ii). Transfer Requirement

Students should have a satisfactory English Proficiency Test score on TOEFL, IELTS or other exams approved by UIUC, and maintain a good standing on academic studies that all term GPAs, overall GPA on UIUC courses are suggested to be above 2.5. In junior year, students will apply for international transfer to be admitted to UIUC and change academic status from non-degree student to degree student in Fall semester of senior year.

2.2 Graduation Requirements for ZJU degree

In addition to the 128-AU requirement listed above, the curriculum leading to the degree of Bachelor of Engineering from ZJU requires students to complete additional **33.5** academic credits of ZJU-required liberal education courses for domestic students and additional seven courses (**23** AUs) in Chinese language and society study for international students. Please refer to section 5 for details.

3. Required Courses

3.1 Orientation and Professional Development

This course introduces students to the opportunities and resources our institute and curriculum can offer you as you work to achieve your career goals. It also provides the skills to work effectively and successfully in the engineering profession.

Course Code	Course Name	Academic Credits
ENG 100	Engineering Orientation	1
CEE 190	Project Based Introduction to Civil Engineering	4
CEE 495	Professional Practice	0
	Total Required AUs:	**5**

3.2 Foundational Mathematics and Science

These courses emphasize the fundamental mathematical and scientific principles upon which the engineering discipline is based.

Course Code	Course Name	Academic Credits
MATH 221	Calculus I	4
MATH 231	Calculus II	3

MATH 241	Calculus III	4
MATH 257	Linear Algebra with Computational Applications	3
MATH 285	Intro Differential Equations	3
CHEM 102	General Chemistry I	3
CHEM 103	General Chemistry Lab I	1
CHEM 104	General Chemistry II	3
CHEM 105	General Chemistry Lab II	1
PHYS 211	University Physics: Mechanics	4
PHYS 212	University Physics: Elec & Mag	4
PHYS 213	University Physics: Thermal Physics	2
	Total Required AUs:	**35**

3.3 Technical Core

These courses stress fundamental concepts and basic laboratory techniques that comprise the common intellectual understanding of mechanical engineering.

Course Code	Course Name	Academic Credits
CS 101	Intro Computing: Engineering & Science	3
SE 101	Engineering Graphics & Design	3
CEE 201	Systems Engineering & Economics	3
CEE 202	Engineering Risk & Uncertainty	3
TAM 211	Statics	3
TAM 212	Introductory Dynamics	3
TAM 251	Introductory Solid Mechanics	3
TAM 335	Introductory Fluid Mechanics	4
	Total Required AUs:	**25**

3.4 Composition

i) These courses teach fundamentals of expository writing.

Course Code	Course Name	Academic Credits

RHET 101	Principles of Writing	4
RHET 102	Principles of Research	4
	Total Required AUs:	**8**

ii) The Advanced Composition requirement is fulfilled by a writing-intensive course beyond basic composition. It is normally taken in the junior or senior year.

Course Code	Course Name	Academic Credits
CEE 300*	Behaviors of Materials	4
	Total Required AUs:	**4**

* Satisfies the Advanced Composition requirement.

4. Elective Courses

4.1 Science Electives

This elective allows the student to gain additional depth in science. The course should be selected according to the requirements and recommendations for the selected primary area of study in civil engineering, which is subject to approval by ZJUI and UIUC.

Select **one (1)** course, 3 AUs, from the following course list.

Course Code	Course Name	Academic Units
ATMS 120	Severe and Hazardous Weather	3
ATMS 303	Synoptic-Dynamic Wea Analysis	4
CHBE 321	Thermodynamics	4
CHEM 232	Elementary Organic Chemistry I	3 or 4
CS 357	Numerical Methods I	3
ECE 110	Introduction to Electronics	3
ECE 205	Electrical and Electronic Circuits	3
ENSU 300	Environmental Sustainability	3
ESE 140	Climate and Global Change	3
ESE 320	Water Planet, Water Crisis	3
ESE 482	Challenges of Sustainability	3
FIN 221	Corporate Finance	3
FIN 230	Introduction to Insurance	3
GEOL 107	Physical Geology	4
GEOL 118	Natural Disasters	3

GEOL 333	Earth Materials and the Env	4
GEOL 380	Environmental Geology	4
GGIS 103	Earth's Physical Systems	4
LAW 301	Introduction to Law	3
MCB 300	Microbiology	3
ME 200	Thermodynamics	3
ME 340	Dynamics of Mechanical Systems	3.5
MSE 201	Phases and Phase Relations	3
MSE 401	Thermodynamics of Materials	3
NPRE 201	Energy Systems	3
NRES 287	Environment and Society	3
NRES 439	Env and Sustainable Dev	3
SE 320	Control Systems	4
SE 400	Engineering Law	3 or 4
STAT 420	Methods of Applied Statistics	3 or 4
UP 205	Ecology & Environmental Sustainability	3
UP 406	Urban Ecology	4
	Total Required AUs:	**3**

4.2 Technical Electives

This course work is designed to give each student a broad background in the areas of civil engineering through the core courses and to allow each student to develop a focused program through advanced technical electives in chosen primary and secondary fields. Briefly, there are two types of civil engineering courses:

(1) core courses

(2) advanced technical courses.

The core courses are prerequisites for all advanced technical courses. The advanced technical courses are subdivided into a primary area of emphasis and a secondary area of emphasis. The core and secondary area courses assure adequate breadth in civil engineering subjects, while the primary area courses allow the student to study a certain subject in greater depth.

Students cannot use an advanced technical course to meet both primary and secondary requirements.

A student must have advanced technical electives as **12 AUs** in the primary area and **6 AUs** in the secondary area**.**

The sum of academic credits of core courses and technical electives must be at least **thirty-four (34)**.

4.2.1 Core Courses

Dual degree students must select at least **five (5)** courses from the following list. Students pursuing only ZJU degree may choose four courses instead.

Course Code	Course Name	Academic units
CEE 300	Behavior of Materials	4
CEE 310	Transportation Engineering	3
CEE 320	Construction Engineering	3
CEE 330	Environmental Engineering	3
CEE 340	Energy and Global Environment	3
CEE 350	Water Resources Engineering	3
CEE 360	Structural Engineering	3
CEE 380	Geotechnical Engineering	3
	Total Required AUs:	**15**

Core courses cannot be used as advanced technical courses.

4.2.2 Advanced Technical Electives

The advanced technical electives are selected to satisfy the requirements of a primary area of emphasis (i.e., a major field within civil engineering) and a secondary area of emphasis (i.e., a minor field within civil engineering). The program must have at least **12 AUs** in the primary area and **6 AUs** in the secondary area.

ZJUI students are recommended to choose one as primary area from the following:

(1) Structural Engineering (STRUCTURES),
(2) Transportation Engineering (TRANSPORTATION),
(3) Water Resources Engineering & Science (WRES).

The courses in the secondary area of emphasis are chosen to complement the primary area and add breadth to the program of study. Courses that make up a secondary area can be taken in one of the above three recommended areas of civil and environmental engineering, but there are also some additional options-CEE Multidisciplinary that give flexibility to the program.

A secondary program **CANNOT** be taken in the same area as the primary. The secondary area requirement is meant to provide the student both with additional breadth and with an additional area of special focus.

I. Structural Engineering

Structural engineering involves the analysis, design, and construction of buildings, dams, bridges, and other types of facilities. A structural engineer designs economical structures that satisfy requirements of safety, utility, and durability; oversees the building of constructed facilities; and investigates the performance of structures that fail to perform as expected. The tools of the structural engineer include physical testing, mathematical modeling, and computer simulation. The structural engineer uses these tools to make decisions that aid in the creation, maintenance, or demolition of constructed facilities. The largest of structures, such as the Golden Gate Bridge, Hoover Dam, and Eiffel Tower, stand as monuments to the engineering achievements of humankind. The smallest of structures, such as thin films that contain computer circuitry, make possible many devices in our technologically sophisticated society.

Primary in Structural Engineering

Science Elective

Required: *(Select one course from recommended list.)*

Recommended: CS 357, ECE 110, ECE 205, GEOL 107, GEOL 118, ME 200

Civil Engineering Core Courses

Required: CEE 300, CEE 360, CEE 380

Recommended: CEE 320

Advanced Technical Courses

Required: CEE 460, CEE 461, **CEE 465***, CEE 470

Recommended: CEE472

**CEE 465-Design of Structural System is the required Integrated Design course*

Secondary in Structural Engineering

Civil Engineering Core Courses

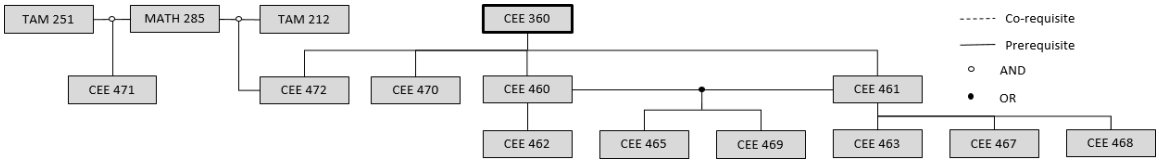
Required: CEE 360

Advanced Technical Courses

Required: CEE 460, CEE 461

Recommended: None

Prerequisite structure for advanced courses in this field:



II. Transportation Engineering

Transportation engineers design, build, operate and maintain all types of facilities for railroads, automobiles, airplanes, and ships. They deal with the physical infrastructure of our transportation system, highways, ports, airports, and other facilities. They are involved in controlling traffic and in developing better transportation systems. Transportation engineers must understand the many factors that affect the long-term performance of the infrastructure from climatic factors imposed on it to the dynamics of the vehicles that use it. Transportation engineers are concerned with the development and operation of our multi-modal transportation system to meet the constantly changing social, economic, geographical, and political needs of society.

Primary in Transportation Engineering

Science Elective

Required:	(Select one course from recommended list.)
Recommended:	ATMS 120, CS 357, ECE 110, ECE 205, GEOL 107, GEOL 118, ME 200, ME 340, MSE 401, SE 320, STAT 420

Civil Engineering Core Courses

Required:	CEE 300, CEE 310
Recommended:	CEE 320, CEE 330, CEE 350, CEE 360, CEE 380

Advanced Technical Courses

Required: Students must select one course from the each of the three Areas and one from the recommended list.

Area 1-Facilities:	CEE 405, CEE 406, CEE 498TIS
Area 2-Systems:	CEE 407, **CEE 415***, CEE 416, CEE 418
Area 3-Railroad:	CEE 408, CEE 409, CEE 410, CEE 411
Recommended:	CEE 401, CEE 405, CEE 406, CEE 408, CEE 409, CEE 410, CEE 411, CEE 412, CEE 415, CEE 416, CEE 417, CEE 418

**CEE 415-Geometric Design of Roads is the required Integrated Design course*

Secondary in Transportation Engineering

Civil Engineering Core Courses

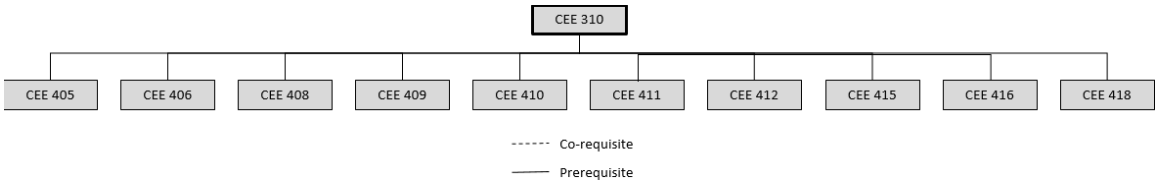
Required:	CEE 310

Advanced Technical Courses

Required: Students must select two courses, each from a different Area.

Area 1-Facilities:	CEE 405, CEE 406, CEE 498TIS
Area 2-Systems:	CEE 407, CEE 415, CEE 416, CEE 418
Area 3-Railroad:	CEE 408, CEE 409, CEE 410, CEE 411, CEE 412
Recommended:	None

Prerequisite structure for advanced courses in this field:



III. Water Resources Engineering & Science

Water Resources Engineering and Science deals with surface and ground water used for human consumption, energy, food, transportation, and recreation, as well as mitigation of floods and other water related hazards. This area comprises the fields of hydrology, hydraulics, and water resources planning. Hydrology is the study of water in the natural environment. Hydraulics studies how to design infrastructure to manage water flow and storage. Water resources planning concerns how to manage water efficiently and economically under hydrologic, hydraulic, economic, and societal constraints. Engineers in this discipline are responsible for the planning, design, operation, and management of facilities for domestic, municipal, industrial and agricultural water supply, drainage, control, and utilization. Such facilities include river channel-dam-reservoir systems for flood control, hydropower, navigation, water supply, and recreation; water distribution networks, sewer systems and culverts for urban water supply and drainage; injection/extraction well systems for remediation of contaminated ground water; and erosion and sediment control structures. Water Resources Engineers are also involved in the planning, operation, and management of ground water, watersheds, and wetlands, as well as the hydrologic environment's response to human impacts and climatic changes.

Primary in Water Resources Engineering and Science

Science Elective

Required:	(Select one course from recommended list.)
Recommended:	ATMS 120, CS 357, GEOL 107, GEOL 118, ME 200

Core Courses

Required:	CEE 300, CEE 350
Recommended:	CEE 310, CEE 320, CEE 330, CEE 340, CEE 360, CEE 380

Advanced Technical Courses

Required:	CEE 452 or **CEE 453***
Recommended:	CEE 432, CEE 433, CEE 434, CEE 437, CEE 450, CEE 451, CEE 452, CEE 453, CEE 457, CEE 458, CEE 459

**CEE 453-Urban Hydrology and Hydraulics is the required Integrated Design course*

Secondary in Water Resources Engineering and Science

Civil Engineering Core Courses

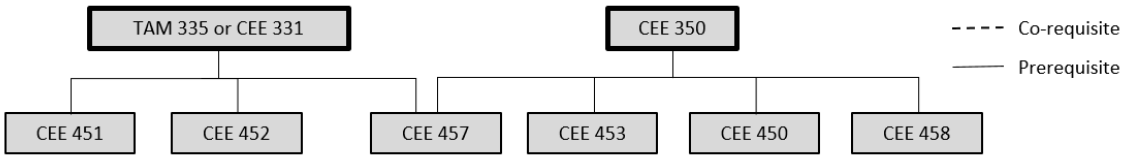
Required: CEE 350

Advanced Technical Courses

Required: Two from the recommended list

Recommended: CEE 432, CEE 433, CEE 434, CEE 450, CEE 451, CEE 452, CEE 453, CEE 457, CEE 458, CEE 459

Prerequisite structure for advanced courses in this field:



IV. The General Civil Engineering Option

The General Option was created to recognize that there are career opportunities in Civil and Environmental Engineering for which great breadth of understanding of all aspects of Civil Engineering has merit over specialization. Some civil and environmental engineers solve problems, which require the interaction with other engineering and non-engineering disciplines. These problems can, and often do, involve multiple teams with diverse expertise and background. Students choosing the General Civil Engineering option are encouraged to discuss with their faculty advisor and/or CEE advising staff.

Requirements of the General Option

Science Electives

Required: One from recommended list

Recommended: GEOL 107, CHEM 222, CHEM 232, ME 200, STAT 400

Civil Engineering Core Courses

Required: CEE 300, CEE 310, CEE 320, CEE 330, CEE 340, CEE 350, CEE 360, CEE 380 (7 of these 8)

Advanced Technical Courses

Required: Option I: Pick no more than one course from each area to satisfy the requirement that the sum of the core and advanced courses be at least 34 AUs.

	Option II: Pick two courses from one area and no more than one course from each of the remaining areas until you reach 34 AUs

Acceptable advanced technical courses:

Construction	CEE 420, CEE 421, CEE 422
Environmental	CEE 437, CEE 440, CEE 446
Geotechnical	CEE 483, CEE 484
Materials	CEE 401
Structures	CEE 460, CEE 461
Transportation	CEE 405, CEE 406, CEE, 408, CEE 409, CEE 410, CEE 411, CEE 412, CEE 415, CEE 416, CEE 417, CEE 418
Water Resources	CEE 452, CEE 453

Notes:

(1) The General Option requires seven of the eight civil engineering core courses, as opposed to just five in the ordinary program. Hence, nearly all of the basic civil engineering disciplines are covered at the core level.

(2) All of the acceptable advanced technical courses are 3 AU courses, except for CEE 401, CEE 415, CEE 453, and CEE 483, which are 4 AU courses. Hence, in most cases this option requires 4 or 5 advanced technical courses.

(3) The acceptable advanced technical courses are listed in numerical order within each area. There is no implication that the first course listed is more appropriate than the second, third, etc.

V. Additional Civil Engineering Secondaries

Secondary Areas in CEE Multidisciplinary

The frontiers of CEE are becoming increasingly multidisciplinary, spanning sub-fields of CEE as well as other engineering and non-engineering fields. The purpose of this secondary is to provide students with interests that do not align with an existing secondary option to develop custom multidisciplinary programs in Mechanical Engineering, Electrical and Computer Engineering.

1) CEE Multidisciplinary – ECE

Required Science Elective – ECE 110
Required Core Course – ECE 210
Recommended Advanced Technical Electives – Two of ECE 310/311, ECE 329, ECE 333, ECE 437 & ECE 486

2) CEE Multidisciplinary – ME

Required Science Elective – ME 200
Required Core Course – ME 270
Required Advanced Technical Electives – Two of ME 340, ME 370, ME 371 & ME 430

4.2.3 List of Some Key CEE Advanced Technical Electives

Course Code	Course Name	Academic Credits
ATMS 120	Severe and Hazardous Weather	3
CEE 401	Concrete Materials	4
CEE 405	Asphalt Materials I	3 or 4
CEE 406	Pavement Design I	3 or 4
CEE 407	Airport Design	3 or 4
CEE 408	Railroad Transportation Engineering	3 or 4
CEE 409	Railroad Track Engineering	3 or 4
CEE 410	Railway Signaling & Control	3 or 4
CEE 411	RR Project Design & Construction	3 or 4
CEE 412	High-Speed Rail Engineering	3 or 4
CEE 415	Geometric Design of Roads	4
CEE 416	Traffic Capacity Analysis	3 or 4
CEE 417	Urban Transportation Planning	4
CEE 418	Public Transportation Systems	3 or 4
CEE 420	Construction Productivity	3 or 4
CEE 421	Construction Planning	3 or 4
CEE 422	Construction Cost Analysis	3 or 4
CEE 424	Sustainable Const Methods	4
CEE 430	Ecological Quality Engineering	2
CEE 432	Stream Ecology	3 or 4
CEE 433	Water Technology and Policy	3 or 4
CEE 434	Environmental Systems I	3
CEE 437	Water Quality Engineering	3
CEE 438	Science & Environmental Policy	3
CEE 440	Fate Cleanup Environ Pollutant	4
CEE 442	Environmental Engineering Principles, Physical	4

CEE 443	Env Eng Principles, Chemical	4
CEE 444	Env Eng Principles, Biological	4
CEE 445	Air Quality Modeling	4
CEE 447	Atmospheric Chemistry	4
CEE 449	Environmental Engineering Lab	3
CEE 450	Surface Hydrology	3
CEE 451	Environmental Fluid Mechanics	3
CEE 452	Hydraulic Analysis and Design	3
CEE 453	Urban Hydrology and Hydraulics	4
CEE 457	Groundwater	3
CEE 458	Water Resources Field Methods	4
CEE 460	Steel Structures I	3
CEE 461	Reinforced Concrete I	3
CEE 465	Design of Structural Systems	3
CEE 470	Structural Analysis	4
CEE 472	Structural Dynamics I	3 or 4
CEE 480	Foundation Engineering	3
CEE 483	Soil Mechanics and Behavior	4
CEE 484	Applied Soil Mechanics	4
CEE 491	Decision and Risk Analysis	3 or 4
CEE 493	Sustainable Design Engineered Technologies	4
CEE 498TIS	Transportation Infrastructure and Smart Mobility	3
CHEM 222	Quantitative Analysis Lecture	2
CHEM 232	Elementary Organic Chemistry I	3 or 4
CS 357	Numerical Methods I	3
ECE 110	Introduction to Electronics	3
ECE 205	Electrical and Electronic Circuits	3
ECE 310	Digital Signal Processing	3
ECE 311	Digital Signal Processing Lab	1
ECE 329	Fields and Waves I	3
ECE 333	Green Electric Energy	3
ECE 437	Sensors and Instrumentation	3

ECE 486	Control Systems	4
GEOL 107	Physical Geology	4
GEOL 118	Natural Disasters	3
ME 200	Thermodynamics	3
ME 340	Dynamics of Mechanical Systems	3.5
ME 370	Mechanical Design I	3
ME 371	Mechanical Design II	3
ME 430	Failure of Engineering Materials	3 or 4
SE 320	Control Systems	4
STAT 420	Methods of Applied Statistics	3 or 4

4.3 General Education

The general education courses develop students' understanding of human culture and society, build skills of inquiry and critical thinking, and lay a foundation for civic engagement and lifelong learning. To satisfy the General Education requirements, students must complete a minimum of **twelve (12)** AUs in general education courses. Students must take all courses for grade and complete sufficient courses in required categories based on the table below:

Required course category	Number of courses	Academic Credits
1) **Humanities & Arts**	Two (2)	6
2) **Social & Behavioral Sciences**[1]	Two (2)	6
Additional Requirements:		
i) Culture Studies[2]		
Western/Comparative Culture(s)	One (1)	3
Non-Western Culture(s)	One (1)	3
U.S. Minority Culture(s)[3]	One (1)	3
ii) Aesthetic Education[4]	One (1)	3
	Total required AUs:	**12**

[1] One of the Social and Behavioral Science courses must be an introductory economics course (ECON 102 or ECON 103).
[2] Proper selections of Social and Behavioral Sciences and in Humanities and the Arts courses will assure that these courses also fulfill Western, non-Western and US Minority Cultures requirements.
[3] Not required if students pursue ZJU degree only.
[4] Required only for ZJU degree.

For courses that fail in multiple categories of general education, the academic units of the course will be counted to satisfy the AU requirements of each category but be counted only once in satisfying the total minimum academic unit requirements (12 AUs) of general education.

4.4 Free Electives

These unrestricted electives give students the opportunity to explore any intellectual area of unique interest. This freedom plays a critical role in helping students define their research specialties. Students are encouraged to take cross-disciplinary courses as free electives.

Students must complete a minimum of **six (6)** AUs in free electives.

5. ZJU Required Liberal Education

These courses introduce modern Chinese history, social development, government policies, etc., and help students improve their English and maintain a healthy lifestyle.

5.1 For domestic students

Domestic students must complete all the courses below to fulfill the graduation requirements for the Bachelor of Engineering Degree at ZJU. These courses can be taught in Chinese. The academic credits in this section are additional to 128 AUs of credits required for UIUC degree.

Course Code	Course Name	Academic Credits
LAW 1002	Ideology, Morality and Rule of Law	3
HIST 2001	Modern Chinese History	3
PHIL 2002	Introduction to the Principle of Marxism	3
PS 2011	Intro.to Mao Thought & Theoretical System of China Socialism	3
PS 3011	General Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era	3
PS 1001	Situation and Policy I	1
PS 2001	Situation and Policy II	1
ENGL 1001	Integrated English I	1.5
ENGL 1002	Integrated English II	1.5
ENGL 2001	Advanced Spoken English I	1.5
ENGL 2002	Advanced Spoken English II	1.5
PE 1001	Physical Education I	1
PE 1002	Physical Education II	1
PE 2001	Physical Education III	1

Course Code	Course Name	Academic Credits
PE 2002	Physical Education IV	1
PE 4021	Physical Education VII-Fitness test and exercise	0.5
MITR 1001	Military Training	2
MITR 2001	Military Theory	2
ST 4001	Senior Thesis	2
	Total Required AUs:	**33.5**

5.2 For International students

International students are required to complete the following courses in Chinese language study and Chinese society to fulfill the graduation requirements along with the above 128 academic credits of courses towards the Bachelor of Engineering Degree at ZJU.

Course Code	Course Name	Academic Credits
CHIN 1001	Chinese I	4
CHIN 1002	Chinese II	4
CHIN 1003	Chinese III	4
CHIN 1004	Chinese IV	4
CHIN 1005	International Chinese Language Proficiency and Testing	2
CULT 2001	China Survey	3
ST 4001	Senior Thesis	2
	Total Required AUs:	**23**

6. Course Schedule Sample by Semester

6.1 First Year-First (Fall) Semester

List of ZJUI/UIUC courses for both domestic and international students

No	Course Code	Course Name	Academic Credits
1	Rhet 101	Principles of Writing	4
2	Chem 102	General Chemistry I	3
3	Chem 103	General Chemistry Lab I	1
4	Math 221	Calculus I	4
5	CEE 190	Project-Based Introduction to CEE	4

6	CS 101	Introduction to Computing: Engineering & Science	3
7	Eng 100	Engineering Orientation	1
		Total:	20

List of ZJU courses for domestic students

No	Course Code	Course Name	Academic Credits
1	MITR 1001	Military Training	2
2	ENGL 1001	Integrated English I	1.5
3	PE 1001	Physical Education I	1
4	PS 1001	Situation and Policy I	

List of ZJU courses for International students

No	Course Code	Course Name	Academic Credits
1	CHIN 1001	Chinese I	4

6.2 First Year-Second (Spring) Semester

List of ZJUI/UIUC courses for both domestic and international students

No	Course Code	Course Name	Academic Credits
1	Rhet 102	Principles of Research	4
2	Math 231	Calculus II	3
3	Phys 211	University Physics: Mechanics	4
4	SE 101	Engineering Graphics & Design	3
5	Chem 104	General Chemistry II	3
6	Chem 105	General Chemistry II Lab	1
		Total:	17

List of ZJU Courses for domestic students

No	Course Code	Course Name	Academic Credits
1	LAW 1002	Ideology, Morality and Rule of Law	3
2	ENGL1002	Integrated English II	1.5
3	PE 1002	Physical Education II	1
4	PS 1001B	Situation and Policies I	1

List of ZJU Courses for international students

No	Course Code	Course Name	Academic Credits
1	CHIN 1002	Chinese II	4

6.3 Second Year-First (Fall) Semester

List of ZJUI/UIUC courses for both domestic and international students

No	Course Code	Course Name	Academic Credits
1	Math 241	Calculus III	4
2	MATH 257	Linear Algebra with Computational Application	3
3	Phys 212	University Physics: Elec& Mag	4
4	TAM 211	Statics	3
5	CEE 201	System Engineering & Economics	3
		Total:	17

List of ZJU Courses for domestic students

No	Course Code	Course Name	Academic Credits
1	PS 2011	Intro.to Mao Thought & Theoretical System of China Socialism	3
2	ENGL 2001	Advanced Spoken English I	1.5
3	PE 2001	Physical Education III	1
4	MITR 2001	Military Theory	2
5	PS 2001	Situation and Policy II	

List of ZJU Courses for international students

No	Course Code	Course Name	Academic Credits
6	CHIN 1003	Chinese III	4
7	CULT 2001	China Survey	3

6.4 Second Year-Second (Spring) Semester

List of ZJUI/UIUC courses for both domestic and international students

No	Course Code	Course Name	Academic Credits

1	Math 285	Intro to Differential Eq	3
2	TAM 212	Introductory Dynamics	3
3	TAM 251	Introductory Solid Mechanics	3
4	CEE 202	Engineering Risk & Uncertainty	3
5	Phys 213	University Physics-Thermal Physics	2
6	GenEd	Liberal Education Elective	3
		Total:	17

List of ZJU Courses for domestic students

No	Course Code	Course Name	Academic Credits
1	HIST 2001	Modern Chinese History	3
2	PHIL 2002	Introduction to the Principle of Marxism	3
3	ENGL 2002	Advanced Spoken English II	1.5
4	PS 2001	Situation and Policy II	
5	PE 2002	Physical Education IV	1

List of ZJU Courses for domestic students

No	Course Code	Course Name	Academic Credits
1	CHIN 1004	Chinese IV	4
2	CHIN 1005	International Chinese Language Proficiency and Testing	2

6.5 Third Year-First (Fall) Semester

No	Course Code	Course Name	Academic Credits
1	TAM 335	Introductory Fluid Mechanics	4
2	CEE 300	Behavior of Materials	4
3	Core2	Civil Engineering Core Course 2	3
4	SciElec	Science Elective	3
5	GenEd	General Education Elective	3
		Total:	17

6.6 Third Year- Second (Spring) Semester

No	Course Code	Course Name	Academic Credits
1	Core3	Civil Engineering Core Course 3	3
2	Core4	Civil Engineering Core Course 3	3
3	Core5	Civil Engineering Core Course 3	3

4	Free Elec	Free Elective	3
5	GenEd	Liberal Education Elective	3
		Total:	15

6.7 Fourth Year-First (Fall) Semester

List of ZJUI/UIUC courses for both domestic and international students

No	Course Code	Course Name	Academic Credits
1	CEE 495	Professional Practice	0
2	ATE1	Advanced Technical Elective 1 (Primary)	3
3	ATE2	Advanced Technical Elective 2 (Primary)	3
4	ATE3	Advanced Technical Elective 3 (Secondary)	3
5	Free Elec	Free Elective	3
6	GenEd	Liberal Education Elective	3
		Total:	15

List of ZJU Courses for domestic students

No	Course Code	Course Name	Academic Credits
1	PS 3011	General Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era	3
2	PS 2001	Situation and Policy II	

6.8 Fourth Year- Second (Spring) Semester

List of ZJUI/UIUC courses for both domestic and international students

No	Course Code	Course Name	Academic Credits
1	ATE4	Advanced Technical Elective 4 (Primary)	3
2	ATE5	Advanced Technical Elective 5 (Primary)	3
3	ATE6	Advanced Technical Elective 6 (Secondary)	3
4	Free Elec	Free Elective	3
5	GenEd	Liberal Education Elective	3
		Total:	15

List of ZJU Courses for domestic students

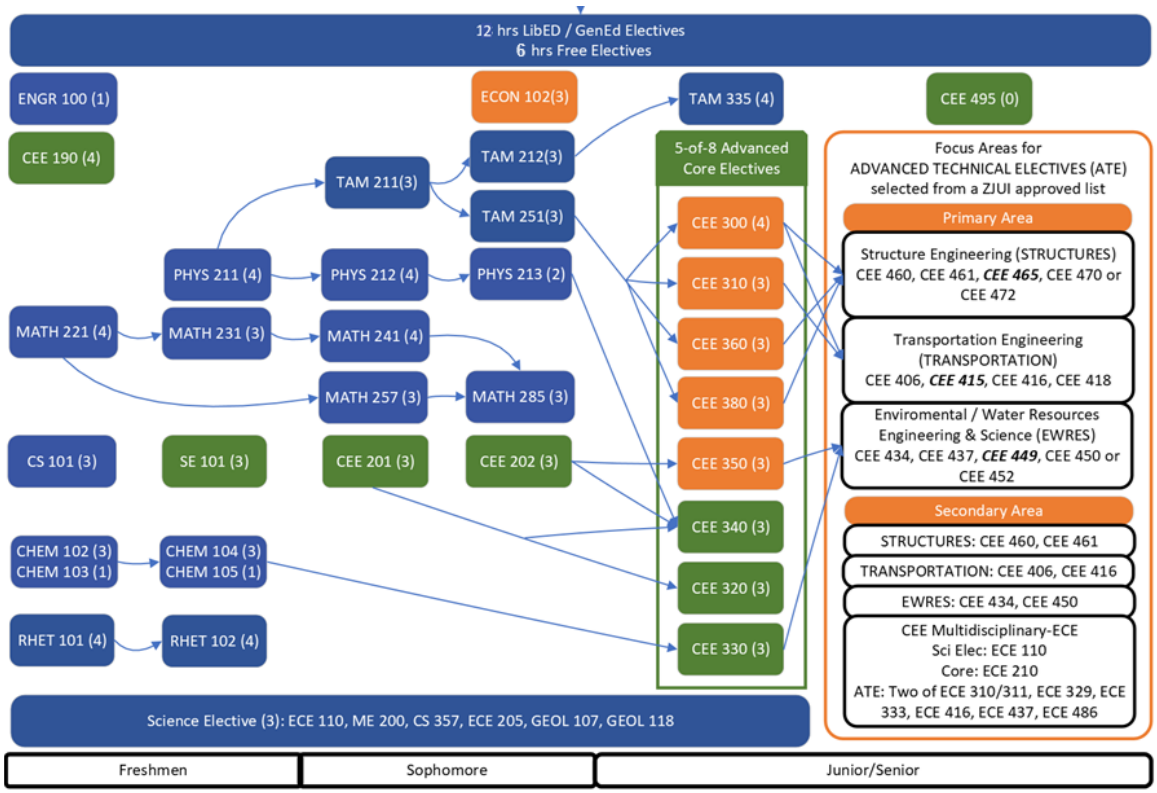
No	Course Code	Course Name	Academic Credits
1	PE 4021	Physical Education VII-Fitness test and exercise	0.5
2	PS 2001F	Situation and Policy Ⅱ	1
3	ST 4001	Senior Thesis	2

List of ZJU Courses for international students

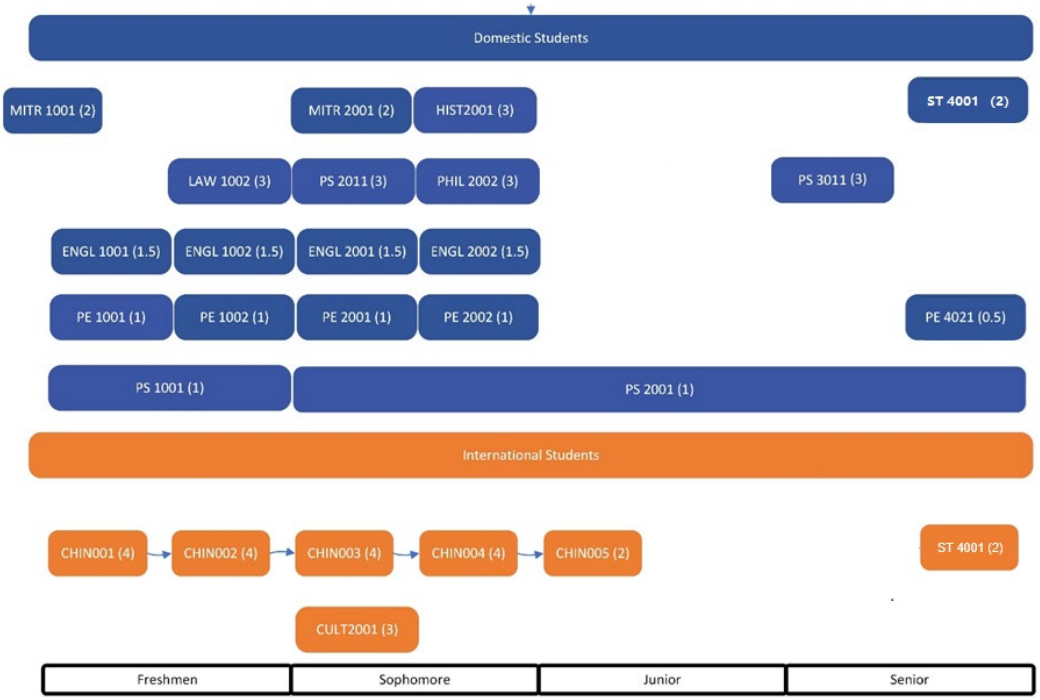
No	Course Code	Course Name	Academic Credits
1	ST 4001	Senior Thesis	2

7. Curriculum Flow Map

The following flow map provides a quick summary of the main features of the civil Engineering curriculum.



ZJU Required Liberal Education Course


Domestic Students
MITR 1001 (2)
MITR 2001 (2)
HIST2001 (3)
ST 4001 (2)
LAW 1002 (3)
PS 2011 (3)
PHIL 2002 (3)
PS 3011 (3)
ENGL 1001 (1.5)
ENGL 1002 (1.5)
ENGL 2001 (1.5)
ENGL 2002 (1.5)
PE 1001 (1)
PE 1002 (1)
PE 2001 (1)
PE 2002 (1)
PE 4021 (0.5)
PS 1001 (1)
PS 2001 (1)
International Students
CHIN001 (4)
CHIN002 (4)
CHIN003 (4)
CHIN004 (4)
CHIN005 (2)
ST 4001 (2)
CULT2001 (3)
Freshmen
Sophomore
Junior
Senior

Appendix A. Integrated Design Courses

An integrated design course is a course that has a design project as part of the course requirements. The following courses meet the conditions of an integrated design course.

Course Code	Course Name
CEE 401	Concrete Materials
CEE 415	Geometric Design of Roads
CEE 421	Construction Planning
CEE 449	Environmental Engineering Lab
CEE 453	Urban Hydrology and Hydraulics
CEE 465	Design of Structural Systems
CEE 484	Applied Soil Mechanics
CEE 493	Sustainable Design Eng Tech

8. Appendix B. Design Content of CEE Courses

The design content of a course is a number representing the number of academic credits of the course that are directly associated with engineering design. The design content of each CEE course in the catalog are listed below:

Course Code	Course Name	Academic Credit
CEE 190	Project Based Introduction to CEE	0.00
CEE 199	Undergraduate Open Seminar	0.00
CEE 201	Systems Engineering & Economics	1.50
CEE 202	Engineering Risk & Uncertainty	1.50
CEE 300	Behavior of Materials	1.00
CEE 310	Transportation Engineering	1.00
CEE 311	Engineering Surveying	1.20
CEE 320	Construction Engineering	1.50
CEE 330	Environmental Engineering	0.50
CEE 331	Fluid Dynamics in the Natural and Built Environment	0.00
CEE 340	Energy and Global Environment	1.00
CEE 350	Water Resources Engineering	1.30
CEE 360	Structural Engineering	0.50
CEE 380	Geotechnical Engineering	1.00
CEE 401	Concrete Materials	1.00
CEE 405	Asphalt Materials, I	2.00
CEE 406	Pavement Design, I	2.50
CEE 407	Airport Design	2.00
CEE 408	Railroad Transportation Engineering	1.00
CEE 409	Railroad Track Engineering	2.00
CEE 410	Railway Signaling and Control	1.00
CEE 411	Railroad Project Design & Construction	2.00

CEE 412	High-Speed Rail Engineering	1.00
CEE 415	Geometric Design of Roads	3.00
CEE 416	Traffic Capacity Analysis	1.50
CEE 417	Urban Transportation Planning	0.00
CEE 418	Public Transportation Systems	1.00
CEE 419	Transportation Economics	0.00
CEE 420	Construction Productivity	2.10
CEE 421	Construction Planning	2.50
CEE 422	Construction Cost Analysis	1.80
CEE 424	Sustainable Const Methods	1.00
CEE 430	Ecological Quality Engineering	1.00
CEE 432	Stream Ecology	1.00
CEE 433	Water Technology and Policy	0.50
CEE 434	Environmental Systems, I	2.00
CEE 435	Public Health Engineering	0.00
CEE 437	Water Quality Engineering	1.50
CEE 438	Science and Environmental Policy	0.25
CEE 440	Fate Cleanup Environ Pollutant	1.00
CEE 441	Air Pollution Sources, Transport, and Control	2.00
CEE 442	Env Eng Principles, Physical	0.50
CEE 443	Env Eng Principles, Chemical	0.00
CEE 444	Env Eng Principles, Biological	0.00
CEE 447	Atmospheric Chemistry	0.00
CEE 449	Environmental Engineering Lab	2.00
CEE 450	Surface Hydrology	0.75
CEE 451	Environmental Fluid Mechanics	0.60
CEE 452	Hydraulic Analysis and Design	2.50

CEE 453	Urban Hydrology and Hydraulics	3.00
CEE 457	Groundwater	0.25
CEE 458	Water Resources Field Methods	1.00
CEE 459	Ecohydraulics	0.80
CEE 460	Steel Structures, I	2.40
CEE 461	Reinforced Concrete, I	2.40
CEE 462	Steel Structures, II	2.40
CEE 463	Reinforced Concrete, II	2.25
CEE 465	Design of Structural Systems	3.00
CEE 467	Masonry Structures	2.40
CEE 468	Prestressed Concrete	2.40
CEE 469	Wood Structures	2.40
CEE 470	Structural Analysis	1.00
CEE 471	Structural Mechanics	0.30
CEE 472	Structural Dynamics	1.00
CEE 473	Wind Effects on Structures	0.00
CEE 474	Mechanics of Additive Manufacturing	0.75
CEE 483	Soil Mechanics and Behavior	1.50
CEE 484	Applied Soil Mechanics	2.50
CEE 490	Computer Methods	0.00
CEE 491	Decision and Risk Analysis	1.50
CEE 492	Data Science for Civil and Environmental Engineering	0.00
CEE 493	Sustainable Design Eng Tech	2.00
CEE 495	Professional Practice	0.00
CEE 497*	Independent Study*	0.00
CEE 498*	Special Topics*	0.00

**Note: Design content in CEE 497 and CEE 498 is by default assigned to be zero.*